베이지안 기계학습
2. Markov chain Monte Carlo (MCMC) I

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Basics of MCMC

*Markov chain Monte Carlo (MCMC)

$$\boldsymbol{\theta}^{(0)} \xrightarrow{q} \boldsymbol{\theta}^{(1)} \xrightarrow{q} \boldsymbol{\theta}^{(2)} \xrightarrow{q} \boldsymbol{\theta}^{(3)} \xrightarrow{q} \boldsymbol{\theta}^{(4)} \xrightarrow{q}$$

- We say $\{\theta^{(k)}\}_{k=1}^{K}$ is a Markov chain if

$$[\theta^{(k+1)} \mid \theta^{(k)}, \ldots, \theta^{(1)}] \overset{d}{=} [\theta^{(k+1)} \mid \theta^{(k)}], \quad \forall k.$$

- To generate a Markov chain, we need
  1. an initial value $\theta^{(0)}$,
  2. “kernel” (or conditional distribution) $q(\theta^{(k+1)} \mid \theta^{(k)})$.

*Markov chain Monte Carlo (MCMC)

- For some function g,

$$
\begin{aligned}
\mathbb{E}\{g(\theta) \mid y\} &= \int g(\theta)\pi(\theta \mid y)d\theta \\
&\approx \frac{1}{K}\sum_{k=1}^{K} g(\theta^{(k)}),
\end{aligned}
$$

where $\theta^{(1)}, \ldots, \theta^{(K)} \overset{iid}{\sim} \pi(\theta \mid y)$.

*Markov chain Monte Carlo (MCMC)

- Under regularity conditions, we can
  1. generate a Markov chain $\{\theta^{(k)}\}_{k=1}^{K}$ such that $\theta^{(k)} \dot{\sim} \pi(\theta \mid x)$;
  2. use the Monte Carlo method

$$\mathbb{E}\{g(\theta) \mid x\} \quad \approx \quad \frac{1}{K}\sum_{k=1}^{K} g(\theta^{(k)}) \ \ \forall \text{large } K,$$

  even though $\{\theta^{(k)}\}_{k=1}^{K}$'s are not i.i.d! (Ergodic theorem)

- An MCMC is a method to obtain such samples, $\{\theta^{(k)}\}_{k=1}^{K}$.
- There are many available MCMC methods, but today we will focus on Gibbs sampler and Metropolis-Hastings algorithm.

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Gibbs sampler

Gibbs sampler

- Gibbs sampler is an MCMC algorithm.
- Suppose we have p-dimensional parameter $\theta = (\theta_1, \ldots, \theta_p)$ and wish to sample θ from the joint posterior $\pi(\theta \mid y)$.
- Gibbs sampler uses full conditional posteriors of each component θ_i, say $\pi(\theta_i \mid \theta_{-i}, y)$.

Gibbs sampler: algorithm

1. Input: initial value $\theta^{(0)} = (\theta_1^{(0)}, \ldots, \theta_p^{(0)})$
2. for $k = 1, \ldots, K$ do

   (1) sample $\theta_1^{(k)} \sim \pi(\theta_1 \mid \theta_2^{(k-1)}, \theta_3^{(k-1)}, \theta_4^{(k-1)}, .., \theta_p^{(k-1)}, y)$

   (2) sample $\theta_2^{(k)} \sim \pi(\theta_2 \mid \theta_1^{(k)}, \theta_3^{(k-1)}, \theta_4^{(k-1)}, \ldots, \theta_p^{(k-1)}, y)$

   (3) sample $\theta_3^{(k)} \sim \pi(\theta_3 \mid \theta_1^{(k)}, \theta_2^{(k)}, \theta_4^{(k-1)}, \ldots, \theta_p^{(k-1)}, y)$

   $\vdots$

   (p) sample $\theta_p^{(k)} \sim \pi(\theta_p \mid \theta_1^{(k)}, \theta_2^{(k)}, \theta_3^{(k)}, \ldots, \theta_{p-1}^{(k)}, y)$

   $\rightarrow$ We have $\theta^{(k)} = (\theta_1^{(k)}, \ldots, \theta_p^{(k)})$.
3. Output: MCMC sample $\{\theta^{(k)}\}_{k=1}^K$

- **Theorem:** For any initial value, $\theta^{(k)} \dot\sim \pi(\theta \mid y)$ for all large k.

next

Gibbs sampler: algorithm

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   $\vdots$

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next

*Example

$$\begin{aligned} Y_1, \ldots, Y_n \mid \mu, \sigma^2 &\overset{iid}{\sim} N(\mu, \sigma^2) \\ \mu &\sim N(\delta, \tau^2) \\ \sigma^2 &\sim IG(\alpha, \beta). \end{aligned}$$

- The joint posterior $\pi(\mu, \sigma^2 \mid y)$ is not available.
- (Full conditional posteriors)

$$\begin{aligned} \mu \mid \sigma^2, y &\sim N\Big(\Big(\frac{1}{\tau^2} + \frac{n}{\sigma^2}\Big)^{-1}\Big(\frac{\delta}{\tau^2} + \frac{n\bar{y}}{\sigma^2}\Big), \ \Big(\frac{1}{\tau^2} + \frac{n}{\sigma^2}\Big)^{-1}\Big), \\ \sigma^2 \mid \mu, y &\sim IG\Big(\frac{n}{2} + \alpha, \ \frac{1}{2}\sum_{i=1}^{n}(y_i - \mu)^2 + \beta\Big). \end{aligned}$$

*Posterior computation

(Full conditional posterior of μ)

$$
\begin{aligned}
\pi(\mu \mid \sigma^2, y) &\propto f(y \mid \mu, \sigma^2)\pi(\mu)\pi(\sigma^2) \\
&\propto f(y \mid \mu, \sigma^2)\pi(\mu) \\
&\propto \exp\Big\{ - \frac{1}{2\sigma^2} \sum_{i=1}^{n} (y_i - \mu)^2 - \frac{1}{2\tau^2}(\mu - \delta)^2 \Big\} \\
&\propto \exp\Big\{ - \frac{1}{2\sigma^2} n(\mu - \bar{y})^2 - \frac{1}{2\tau^2}(\mu - \delta)^2 \Big\} \\
&\propto \exp\Big[- \frac{1}{2}\Big\{\Big(\frac{n}{\sigma^2} + \frac{1}{\tau^2}\Big)\mu^2 - 2\Big(\frac{n\bar{y}}{\sigma^2} + \frac{\delta}{\tau^2}\Big)\mu\Big\}\Big] \\
&\overset{(*)}{\propto} \text{pdf of } N\bigg(\Big(\frac{1}{\tau^2} + \frac{n}{\sigma^2}\Big)^{-1}\Big(\frac{\delta}{\tau^2} + \frac{n\bar{y}}{\sigma^2}\Big),\ \Big(\frac{1}{\tau^2} + \frac{n}{\sigma^2}\Big)^{-1}\bigg)
\end{aligned}
$$

$(*)$ Note that $N(x \mid \mu_0, \sigma_0^2) \propto \exp\big\{ - \frac{1}{2}\big(\frac{1}{\sigma_0^2} x^2 - 2 \frac{\mu_0}{\sigma_0^2} x \big) \big\}$.

*Posterior computation

(Full conditional posterior of σ^2)

$$\begin{aligned}
\pi(\sigma^2 \mid \mu, y) &\propto f(y \mid \mu, \sigma^2)\pi(\mu)\pi(\sigma^2) \\
&\propto f(y \mid \mu, \sigma^2)\pi(\sigma^2) \\
&\propto (\sigma^2)^{-\frac{n}{2}} \exp\Big\{-\frac{1}{2\sigma^2}\sum_{i=1}^{n}(y_i-\mu)^2\Big\}(\sigma^2)^{-a-1}\exp\Big(-\frac{\beta}{\sigma^2}\Big) \\
&= (\sigma^2)^{-(\frac{n}{2}+a)-1}\exp\Big[-\frac{1}{\sigma^2}\Big\{\frac{1}{2}\sum_{i=1}^{n}(y_i-\mu)^2+\beta\Big\}\Big] \\
&\overset{(*)}{\propto} \text{pdf of } IG\Big(\frac{n}{2}+\alpha,\ \frac{1}{2}\sum_{i=1}^{n}(y_i-\mu)^2+\beta\Big)
\end{aligned}$$

$(*)$ Note that $IG(x \mid a, b) \propto x^{-a-1}\exp\left(-\frac{b}{x}\right)$.

*Gibbs sampler

1. Input: initial value $(\mu^{(0)}, {\sigma^2}^{(0)})$
2. for $k = 1, \ldots, K$ do
   (1) sample $\mu^{(k)} \sim \pi(\mu \mid {\sigma^2}^{(k-1)}, y)$
   (2) sample ${\sigma^2}^{(k)} \sim \pi(\sigma^2 \mid \mu^{(k)}, y)$
3. Output: MCMC sample $\{(\mu^{(k)}, {\sigma^2}^{(k)})\}_{k=1}^{K}$

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Metropolis-Hastings (MH) algorithm

Metropolis-Hastings: motivation

- To use the Gibbs sampler, full conditional posteriors, $\pi(\theta_i \mid \theta_{-i}, y)$ for $i = 1, \ldots, p$, should be calculated.
- Suppose we want to sample from the posterior $\pi(\theta \mid y)$.
- Suppose we have a kernel $q(\theta^{(k+1)} \mid \theta^{(k)})$ that is easy to generate samples and

$$supp(\pi(\cdot \mid y)) \subset \bigcup_{\theta^{(k)} \in supp(\pi(\cdot|y))} supp\{q(\cdot \mid \theta^{(k)})\}.$$

- The MH algorithm is an MCMC method that obtains posterior samples using a kernel q satisfying the above condition.

Metropolis-Hastings algorithm

Set the initial value $\theta^{(0)}$ and repeat the following steps for $k = 1, 2, \ldots$:

1. Generate θ^* from
$$\theta^* \sim q(\cdot \mid \theta^{(k-1)}).$$

2. Calculate the acceptance probability:
$$\alpha = \min\left\{\frac{\pi(\theta^* \mid y)}{\pi(\theta^{(k-1)} \mid y)} \frac{q(\theta^{(k-1)} \mid \theta^*)}{q(\theta^* \mid \theta^{(k-1)})},\ 1\right\}.$$

3. Accept or reject θ^*:
$$\theta^{(k)} = \begin{cases} \theta^* & \text{with probability} \quad \alpha, \\ \theta^{(k-1)} & \text{with probability} \quad 1 - \alpha. \end{cases}$$

Theorem: For any initial value, $\theta^{(k)} \,\dot\sim\, \pi(\theta \mid y)$ for all large k.

next

Metropolis-Hastings algorithm

Set the initial value $\theta^{(0)}$ and repeat the following steps for $k = 1, 2, \ldots$:

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Theorem: For any initial value, $\theta^{(k)} \dot\sim \pi(\theta \mid y)$ for all large k.

next

Remark

- It is known that the invariant distribution of the Metropolis-Hastings (MH) algorithm is $\pi(\theta \mid y)$.
- Note that $\pi(\theta \mid y)$ can be known up to a constant, so we don't need to calculate the normalizing constant,

$$f(y) = \int f(y \mid \theta)\pi(\theta)d\theta.$$

- The kernel q determines performance of the algorithm.

*Random walk Metropolis

- If we set q as follows with some symmetric density g,

$$q(\cdot \mid \theta^{(k-1)}) = g(\cdot - \theta^{(k-1)}),$$

  we call it the random walk Metropolis algorithm.

- For example,

$$\begin{aligned} & q(\cdot \mid \theta^{(k-1)}) &=& N(\theta^{(k-1)}, \delta^2) \\ \text{or} \quad & q(\cdot \mid \theta^{(k-1)}) &=& Unif(\theta^{(k-1)} - \delta, \theta^{(k-1)} + \delta) \end{aligned}$$

  for some $\delta > 0$.

*Variants: MH within Gibbs (Hybrid Gibbs sampling)

- In fact, the Gibbs sampler is a special case of the MH algorithm.
- We can combine Gibbs and MH in the obvious way: sampling directly from full conditional when possible and MH otherwise.

Remark: diagnostics using trace plots

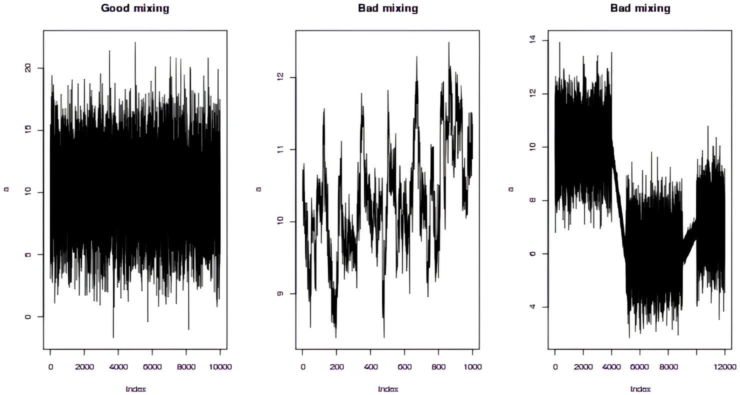

Figure: Good mixing (left) and bad mixing (middle and right) (from https://www.flutterbys.com.au/stats/tut/tut4.3.html).

Remark: diagnostics using trace plots

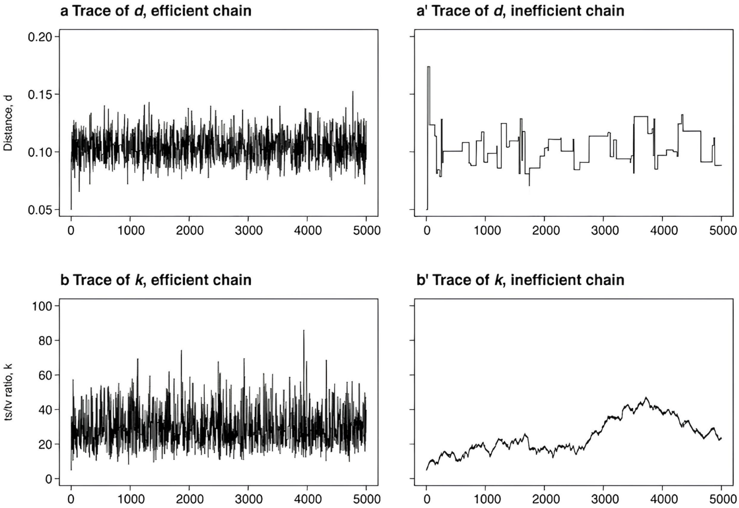

Figure: Good mixing (left column) and bad mixing (right column) (from Nascimento et al. (2017) A biologist's guide to Bayesian phylogenetic analysis).

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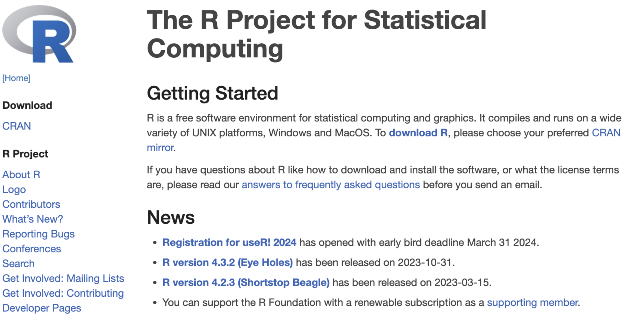
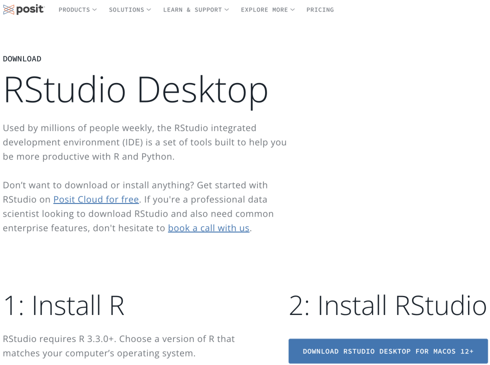
Exercise using R

Install R



1. Go to the R website (`https://www.r-project.org`) and select Download - CRAN.
2. Choose a CRAN mirror close to your physical location for faster downloads.
3. Install R based on your computer's operating system (OS).

Install RStudio



1. Visit the RStudio website (`https://posit.co/download/rstudio-desktop`).
2. Install RStudio based on your computer's OS.

National Football League (NFL) data[1]

$$X_i \mid \lambda_i \overset{ind.}{\sim} Poisson(N\lambda_i), \quad i = 1, \dots, 4$$

- X_i: the number of concussions in the ith year $(i \to 2011 + i)$. We observed $X_1 = 171, X_2 = 152, X_3 = 123$ and $X_4 = 199$.
- $N = 256$: the number of games per season.
- λ_i: the concussion rate per game.

[1] Reich and Ghosh (2021) Bayesian Statistical Methods, CRC Press.

NFL data: prior

$$\begin{aligned} \lambda_i \mid \gamma &\overset{ind.}{\sim} Gamma(1, \gamma), \quad i = 1, \ldots, 4, \\ \gamma &\sim Gamma(a, b), \quad a, b > 0 \end{aligned}$$

- Concussion rates for each year may not be independent of each other.
- The above prior enables to analyze all years simultaneously.
- Parameters to infer: $\theta := (\lambda_1, \ldots, \lambda_4, \gamma)$.

next

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\end{aligned}
$$

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- The above prior enables to analyze all years simultaneously.
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next

- Posterior:

$$\pi(\theta \mid x) \quad \propto \quad \left\{ \prod_{i=1}^{4} f(x_i \mid \lambda_i) \pi(\lambda_i \mid \gamma) \right\} \pi(\gamma)$$

- The joint posterior of $\theta = (\lambda_1, \ldots, \lambda_4, \gamma)$ does not have a closed form.
- For posterior inference, we need MCMC methods.

NFL data: 1. Gibbs sampler

$$
\begin{aligned}
X_i \mid \lambda_i &\overset{ind.}{\sim} Poisson(N\lambda_i), \quad i = 1, \ldots, 4, \\
\lambda_i \mid \gamma &\overset{ind.}{\sim} Gamma(1, \gamma), \quad i = 1, \ldots, 4, \\
\gamma &\sim Gamma(a, b), \quad a, b > 0
\end{aligned}
$$

- We can calculate the conditional posteriors as follows:

$$
\begin{aligned}
\lambda_i \mid \text{rest} &\overset{ind.}{\sim} Gamma(X_i + 1, N + \gamma), \quad i = 1, \ldots, 4, \\
\gamma \mid \text{rest} &\sim Gamma(4 + a, \sum_{i=1}^{4} \lambda_i + b).
\end{aligned}
$$

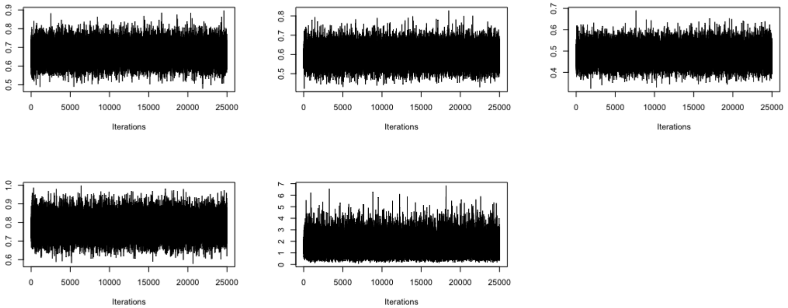
R code

```
# Load data
Y = c(171, 152, 123, 199)
n = 4
N = 256

# Create an empty matrix for the MCMC samples
S = 25000
samples = matrix(NA, S, 5)
colnames(samples) = c("lam1", "lam2", "lam3", "lam4", "gamma")

# Initial values
lambda = Y/N
gamma = 1/mean(lambda)

# priors: lambda[i] | gamma ~ Gamma(1, gamma), gamma ~ IG(a,b)
a = 0.1;  b = 0.1
```

R code: Gibbs sampler

```
# Gibbs sampler
for(s in 1:S){
  for(i in 1:n){
    lambda[i] = rgamma(1, Y[i]+1, N+gamma)
  }
  gamma = rgamma(1, 4+a, sum(lambda)+b)
  samples[s,] = c(lambda, gamma)
}
```

R code: trace plots



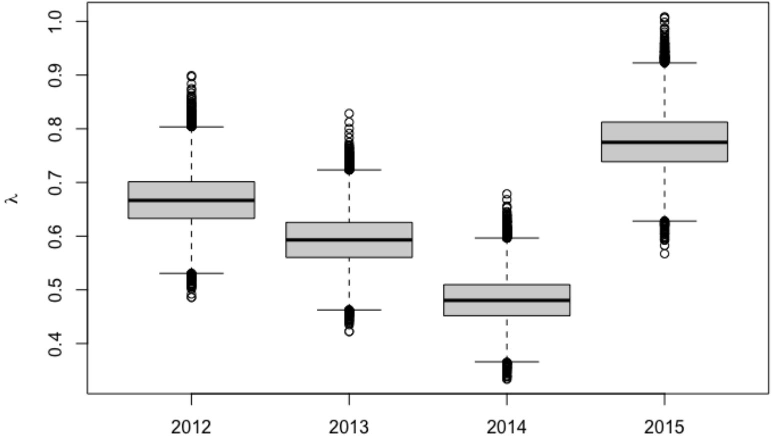

```
library(coda)
par(mfrow=c(2,3))
for(k in 1:4){
  traceplot(as.mcmc(samples[,k]))
}
traceplot(as.mcmc(samples[,5]))
par(mfrow=c(1,1))
```

R code: posterior summary

```
> apply(samples, 2, mean)
     lam1      lam2      lam3      lam4     gamma
0.6678729 0.5938455 0.4815804 0.7763386 1.5716568

> apply(samples, 2, quantile, c(0.025, 0.500, 0.975))
           lam1      lam2      lam3      lam4     gamma
2.5%  0.5729490 0.5032394 0.4013322 0.6728392 0.4497384
50%   0.6665507 0.5929545 0.4803125 0.7747212 1.4394028
97.5% 0.7708054 0.6924060 0.5706073 0.8886379 3.4333161
```

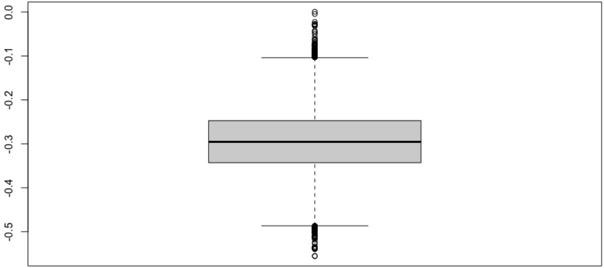
R code: boxplot

```
# Create boxplot for lambda values
boxplot(samples[, 1:4], ylab = expression(lambda), names = 2012:2015)
```



R code: boxplot for $\lambda_3 - \lambda_4$

```
boxplot(samples[,3]-samples[,4])
```




```
> quantile(samples[,3]-samples[,4], probs = c(0.025, 0.975))
      2.5%      97.5%
-0.4327491 -0.1589710
```

NFL data: 2. MH algorithm

$$
\begin{aligned}
X_i \mid \lambda_i &\overset{ind.}{\sim} Poisson(N\lambda_i), \quad i = 1, \ldots, 4, \\
\lambda_i \mid \gamma &\overset{ind.}{\sim} Gamma(1, \gamma), \quad i = 1, \ldots, 4, \\
\gamma &\sim Gamma(a, b), \quad a, b > 0
\end{aligned}
$$

- For each parameter, we consider the following kernel:

$$
\begin{aligned}
q(\lambda_i^* \mid \lambda_i^{(k-1)}) &= Gamma(\lambda_i^* \mid 1, 1), \quad i = 1, \ldots, 4, \\
q(\gamma^* \mid \gamma^{(k-1)}) &= Gamma(\gamma^* \mid 1, 1).
\end{aligned}
$$

R code

```r
# Load data
Y = c(171, 152, 123, 199)
n = 4
N = 256

# Create an empty matrix for the MCMC samples
S = 25000
samples = matrix(NA, S, 5)
colnames(samples) = c("lam1", "lam2", "lam3", "lam4", "gamma")

# Initial values
lambda = Y / N
gamma = 1 / mean(lambda)

# priors: lambda[i] | gamma ~ Gamma(1, gamma), gamma ~ IG(a, b)
a = 0.1
b = 0.1
```

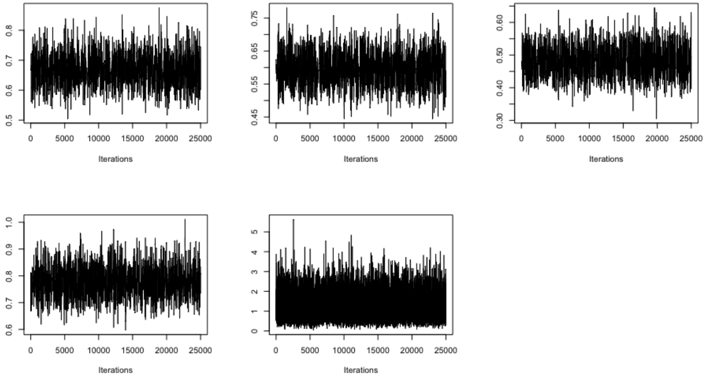
R code

```
# log posterior (up to constant)
log_post <- function(Y, N, lambda, gamma) {
  like = sum(dpois(Y, N * lambda, log = TRUE))
  prior = sum(dgamma(lambda, 1, gamma, log = TRUE)) +
    dgamma(gamma, a, b, log = TRUE)
  post = like + prior
  return(post)
}
```

R code: MH algorithm

```r
for (s in 1:S) {
  for (j in 1:n) {
    can = lambda
    can[j] = rgamma(n = 1, 1, 1)
    logR = log_post(Y, N, can, gamma) - log_post(Y, N, lambda, gamma) -
      dgamma(can[j], 1, 1) + dgamma(lambda[j], 1, 1)
    if (log(runif(1)) < logR) lambda <- can
  }

  can = rgamma(n = 1, 1, 1)
  logR = log_post(Y, N, lambda, can) - log_post(Y, N, lambda, gamma) -
    dgamma(can, 1, 1) + dgamma(gamma, 1, 1)
  if (log(runif(1)) < logR) gamma <- can

  samples[s, ] = c(lambda, gamma)
}
```

R code: trace plots




```
par(mfrow=c(2,3))
for(k in 1:4){
  traceplot(as.mcmc(samples[,k]))
}
traceplot(as.mcmc(samples[,5]))
par(mfrow=c(1,1))
```

R code: posterior summary

```r
# Posterior mean
apply(samples, 2, mean)
# Result:
#      lam1      lam2      lam3      lam4     gamma
# 0.6671055 0.5946340 0.4809925 0.7760812 1.2206463

# Quantiles for 2.5%, 50%, and 97.5%
apply(samples, 2, quantile, c(0.025, 0.500, 0.975))
# Result:
#            lam1      lam2      lam3      lam4     gamma
# 2.5%  0.5693124 0.5062016 0.3951966 0.6743690 0.3601198
# 50%   0.6651682 0.5948847 0.4795900 0.7737184 1.1287282
# 97.5% 0.7725473 0.6897220 0.5675323 0.8915849 2.6372467
```

R code: boxplot

```
# Create boxplot for lambda values
boxplot(samples[, 1:4], ylab = expression(lambda), names = 2012:2015)
```

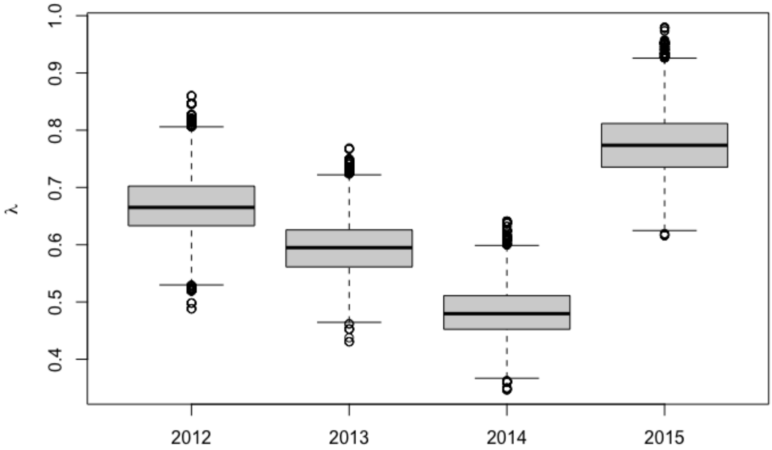


Table of Contents

Remark and next week

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- However, they have certain drawbacks.
  - The Gibbs sampler requires the calculation of full conditional posteriors.
  - The MH algorithm faces challenges related to the choice of the kernel.
- When there is substantial correlation among the parameters, both algorithms may struggle to thoroughly explore the posterior and risk getting trapped in narrow regions.

  $\longrightarrow$ Hamiltonian Monte Carlo

next

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next

Next week

Markov chain Monte Carlo (MCMC) II